

UPENDO DISPENSARY MANAGEMENT SYSTEM

Complete System Documentation with NHIF Integration

Final Budget: Tsh 2,300,000

Facility: Upendo Dispensary – Tanzania

1. System Overview

The **Upendo Dispensary Management System** is a complete, standalone software solution designed specifically for a Tanzanian dispensary. It manages all daily operations digitally, from patient check-in to billing, prescriptions, and reporting.

The system includes a dedicated **NHIF Integration Module** that connects to the National Health Insurance Fund (NHIF) API for real-time member verification and claim submission.

All modules are fully functional, integrated, and ready for use immediately after deployment.

2. Complete List of Modules in the System

The system is built with **10 core modules** that cover every aspect of dispensary operations.

No.	Module Name	Purpose
1	Patient Management Module	Register, search, and manage all patients
2	Consultation & Prescription Module	Digital diagnosis & medicine prescribing
3	Pharmacy & Dispensing Module	Manage stock, dispense medicines
4	Billing & Payment Module	Handle cash & insurance payments
5	NHIF Integration Module	Verify members & submit claims via API
6	Insurance Claims Management Module	Track all NHIF claims status
7	User & Role Management Module	Control access for staff
8	Reports & Analytics Module	Generate business & clinical reports
9	Admin Dashboard Module	Overview of all activities
10	System Settings & Backup Module	Configure facility, backup data

3. Detailed Explanation of Each Module

Module 1 – Patient Management Module

What it does:

- Register new patients with unique **UPD-Number**
- Store: Full name, age, gender, village, phone number, NHIF number (if any), next of kin
- Search patient by: UPD-number, phone number, or NHIF number
- View patient history (past visits, prescriptions, payments)

Key features:

- Auto-complete for returning patients
- Mark patient as **NHIF member** or **Cash patient**
- Edit patient details
- Print patient registration slip

Module 2 – Consultation & Prescription Module

What it does:

- Doctor/Nurse selects a patient and records:
 - Symptoms
 - Vital signs (temperature, blood pressure, weight)
 - Diagnosis (free text or select from common Tanzanian diseases list – Malaria, UTI, RTI, etc.)
- Write prescription from medicine list

Key features:

- Medicine database includes: name, strength, form (tablet/syrup/injection), price
- Prescription fields: dose, frequency (e.g., BD, TDS), duration (days)
- Prescription saved and sent automatically to Pharmacy module
- Print consultation note for patient

Module 3 – Pharmacy & Dispensing Module

What it does:

- Pharmacist sees all pending prescriptions
- Dispense medicines one by one
- Reduce stock quantity automatically
- Alert if medicine stock is low (e.g., <10 items left)

Key features:

- View prescription details before dispensing
- Record “out of stock” and suggest alternative
- Mark prescription as **Dispensed**
- Print medicine label + instruction slip in Swahili/English

Module 4 – Billing & Payment Module

What it does:

- Automatically calculate total bill:
 - Consultation fee (fixed)
 - Medicine costs
 - Any procedures (injection, dressing, etc.)
- Accept two payment types:
 - **Cash** – record amount paid, print receipt
 - **NHIF** – calculate co-pay (if any) and mark for claim submission

Key features:

- Partial payment support
- Discount option (with admin password)
- Invoice number generation

- Daily cash closing report

Module 5 – NHIF Integration Module (*Detailed separately below*)

This module connects directly to NHIF API (or simulated API if NHIF sandbox not yet available). It handles:

1. **Member Verification** – Check if NHIF number is active before consultation
2. **Claim Submission** – Send outpatient claim with diagnosis and medicines
3. **Claim Status Check** – Know if NHIF has paid or rejected

Module 6 – Insurance Claims Management Module

What it does:

- Keep a complete log of all NHIF claims submitted
- For each claim store:
 - Claim ID (from NHIF)
 - Patient name & NHIF number
 - Total amount claimed
 - Status: *Pending / Approved / Rejected / Paid*
 - Date submitted
 - If rejected: reason from NHIF

Key features:

- Filter claims by date range or status
- Re-submit a rejected claim after correction
- Export claims list to Excel for reconciliation

Module 7 – User & Role Management Module

What it does:

- Create system users (staff)
- Assign roles with specific permissions

Roles included:

Role	Permissions
Admin	Everything – users, settings, backups, reports
Doctor/Nurse	Consultation, prescription, view patient history
Pharmacist	Dispense medicines, view stock, submit NHIF claims
Receptionist	Register patients, verify NHIF, cash payment
NHIF Officer	View claims, re-submit failed claims, run NHIF reports
Accountant	View billing reports, cash summary, print receipts

Key features:

- Password change on first login
- Inactive user lock after 5 failed attempts
- Activity log (who did what)

Module 8 – Reports & Analytics Module

Reports included:

Report Name	Description
Daily Revenue Report	Cash vs NHIF totals for the day
Patient Visit Report	Number of patients per day/week/month
NHIF Claims Report	All claims, status, amounts
Medicine Usage Report	Most dispensed medicines
Low Stock Report	Medicines below minimum quantity
Financial Summary	Monthly income by payment type

Export formats: PDF, Excel, Print

Visuals: Bar charts and pie charts (optional but included)

Module 9 – Admin Dashboard Module

What it shows on one screen:

- Today's total patients (cash + NHIF)
- Pending prescriptions (not yet dispensed)
- Pending NHIF claims (not yet submitted)
- Low stock alerts
- Quick buttons: New patient, New consultation, Submit claims

Design: Clean, large fonts, touch-friendly for tablet use at reception.

Module 10 – System Settings & Backup Module

What it does:

- Configure dispensary details: name, address, phone, logo
- Set consultation fee (e.g., Tsh 5,000)
- Manage medicine list (add/edit/delete medicines & prices)
- NHIF API settings: endpoint URL, API key, timeout
- Backup database (one-click backup to local folder or cloud)
- Restore from backup
- View system logs (errors, logins, NHIF API calls)

4. NHIF Integration Module – Full Technical Explanation

This is **not a separate system** but a built-in module that connects to NHIF.

4.1 What the NHIF Module Does

Function	Description
Verify Member	Before treating a patient, receptionist enters NHIF number → system calls NHIF API → returns “Active” or “Invalid”
Submit Claim Automatically	After consultation and dispensing, system creates a claim (diagnosis, medicines, costs) → sends to NHIF API → stores response
Manual Claim Resubmission	If a claim fails, NHIF officer can correct and resend
Claim Status Tracking	System checks status daily (if NHIF provides status API) or allows manual status update

4.2 Workflow Step-by-Step

1. **Patient arrives** → Receptionist selects “NHIF Patient” → enters NHIF number.
2. System calls verify_member API → shows **Green: Active** or **Red: Invalid**.
3. If active, patient goes to consultation.
4. Doctor prescribes medicines.
5. Pharmacist dispenses.
6. System calculates:
 - Total bill = Tsh X
 - NHIF covers capitation amount (e.g., Tsh 10,000)
 - Patient pays co-pay (if any, e.g., Tsh 0 or Tsh 5,000)
7. System creates claim JSON:

json

{

"nhif_number": "12345678A",

"patient_name": "John Mkono",

```
"visit_date": "2026-04-29",  
"diagnosis": "Malaria",  
"medicines": [{"name": "Artemether", "qty": 6, "price": 3000}],  
"total_claim": 18000  
}
```

8. System sends to NHIF endpoint → gets claim_id.
9. Staff prints patient receipt + claim acknowledgement.
10. Daily report shows all claims sent.

4.3 UI for NHIF Module

- **Verification screen:** Large badge showing NHIF status.
- **Claim submission popup:** Button “Submit to NHIF” with confirmation.
- **Claims log table:** Columns – Date, NHIF No, Amount, Status (Pending/Approved/Rejected).
- **Rejection reason viewer:** Shows why claim failed (e.g., “Invalid diagnosis code”).

4.4 NHIF API Handling (Technical)

- **Protocol:** HTTPS REST API (JSON)
- **Authentication:** API key + facility code (provided by NHIF after registration)
- **Timeouts:** 30 seconds → if timeout, claim saved as “Pending retry”
- **Offline mode:** If internet is down, claims saved locally and submitted later automatically when connection returns.

5. Security & Data Protection

Included in the Tsh 2.3M:

- All passwords hashed (bcrypt)
- HTTPS support (SSL ready)
- Role-based access control (no one sees what they shouldn't)
- Daily automatic database backup

- No hardcoded API keys – stored in encrypted config file
- Log of all NHIF API requests for audit

6. What You Get After Payment (Final Deliverables)

1. **Complete source code** (PHP/MySQL or Python/Django – client choice)
2. **Installation file** (one-click setup on Windows/Linux)
3. **User manual** (PDF – English & Swahili)
4. **Training session** (half day at Upendo Dispensary)
5. **30 days free support** (WhatsApp/Email)
6. **Database backup script** included

7. Final Cost: Tsh 2,300,000 (All Modules Included)

Component	Cost (Tsh)
All 10 modules (Patient to Reports)	1,200,000
NHIF Integration (API + claim logic)	600,000
Admin Dashboard & User Roles	250,000
Security & Backups	150,000
Reports & Export (Excel/PDF)	100,000
Documentation + Training	100,000
30 days post-deployment support	100,000
TOTAL	2,300,000

No extra fees. No monthly license. You own the system.

8. Conclusion

This **Upendo Dispensary Management System** is a **complete, ready-to-use solution** with all modules working together. NHIF integration is fully built-in – not an add-on. The system

matches Tanzanian dispensary workflows and gives you full control over patient data, billing, and insurance claims.

To proceed: Provide NHIF facility credentials (if already registered) or we will simulate the API until NHIF gives real access.

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